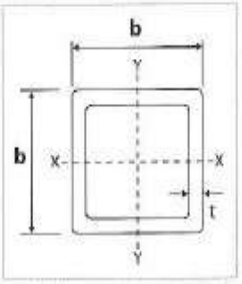
6c (i) Cold Formed Square Hollow Sections

(American Standard)
(Dimension and Properties in accordance to
ASTM A - 500 Grade A and B / Manufacturer's Standard)



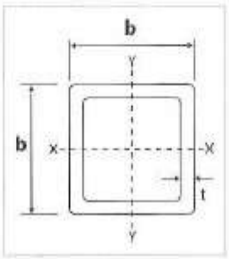
Square Hollow Sections

ASTM A - 500

Nominal Size		Wall Thickness		Calculated Weight			Area	Moment Of Inertia		Section Modulus		Radius Of Gyration		Plastic Modulus	
b x b		t					A	Ix	Iy	Zx	Zy	ix	iy	Sx	Sy
mm	in	mm	in	Kg/m	Kg/6m	Kg/ft	cm^2	cm^4	cm^4	cm^3	cm^3	cm	cm	cm^3	cm^3
20 x 20	-	1.5	0.059	0.83	4.96	0.25	1.05	0.58	0.58	0.58	0.58	0.74	0.74	0.72	0.72
25 x 25	1 x 1	1.5	0.059	1.06	6.37	0.32	1.35	1.22	1.22	0.97	0.97	0.95	0.95	1.17	1.17
		1.9	0.075	1.31	7.83	0.40	1.66	1.44	1.44	1.15	1.15	0.93	0.93	1.41	1.41
		2.3	0.091	1.53	9.19	0.47	1.95	1.61	1.61	1.29	1.29	0.91	0.91	1.62	1.62
		3.0	0.120	1.89	11.34	0.58	2.41	1.84	1.84	1.47	1.47	0.87	0.87	1.91	1.91
30 x 30		1.5	0.059	1.30	7.78	0.40	1.65	2.20	2.20	1.46	1.46	1.15	1.15	1.74	1.74
		1.9	0.075	1.60	9.62	0.49	2.04	2.62	2.62	1.75	1.75	1.13	1.13	2.12	2.12
		2.3	0.091	1.89	11.36	0.58	2.41	2.99	2.99	2.00	2.00	1.11	1.11	2.45	2.45
		3.0	0.120	2.36	14.17	0.72	3.01	3.50	3.50	2.34	2.34	1.08	1.08	2.96	2.96
32 x 32	1 1/4 x 1 1/4	1.5	0.059	1.39	8.35	0.42	1.77	2.70	2.70	1.69	1.69	1.23	1.23	2.00	2.00
		1.9	0.075	1.72	10.34	0.53	2.19	3.24	3.24	2.03	2.03	1.22	1.22	2.44	2.44
		2.3	0.091	2.04	12.23	0.62	2.60	3.71	3.71	2.32	2.32	1.20	1.20	2.84	2.84
		3.0	0.120	2.55	15.30	0.78	3.25	4.38	4.38	2.74	2.74	1.16	1.16	3.44	3.44
35 x 35		1.5	0.059	1.53	9.19	0.47	1.95	3.60	3.60	2.05	2.05	1.36	1.36	2.43	2.43
		1.9	0.075	1.90	11.41	0.58	2.42	4.34	4.34	2.48	2.48	1.34	1.34	2.97	2.97
		2.3	0.091	2.25	13.53	0.69	2.87	4.99	4.99	2.85	2.85	1.32	1.32	3.46	3.46
		3.0	0.120	2.83	16.99	0.86	3.61	5.95	5.95	3.40	3.40	1.28	1.28	4.23	4.23
38 x 38	1 1/2 x 1 1/2	1.5	0.059	1.67	10.04	0.51	2.13	4.67	4.67	2.46	2.46	1.48	1.48	2.89	2.89
		1.9	0.075	2.08	12.48	0.63	2.65	5.65	5.65	2.98	2.98	1.46	1.46	3.54	3.54
		2.3	0.091	2.47	14.83	0.75	3.15	6.54	6.54	3.44	3.44	1.44	1.44	4.15	4.15
		3.0	0.120	3.12	18.69	0.95	3.97	7.85	7.85	4.13	4.13	1.41	1.41	5.10	5.10
40 x 40		1.5	0.059	1.77	10.61	0.54	2.25	5.49	5.49	2.75	2.75	1.56	1.56	3.22	3.22
		1.9	0.075	2.20	13.20	0.67	2.80	6.66	6.66	3.33	3.33	1.54	1.54	3.96	3.96
		2.3	0.091	2.62	15.69	0.80	3.33	7.73	7.73	3.86	3.86	1.52	1.52	4.64	4.64
		3.0	0.120	3.30	19.82	1.01	4.21	9.32	9.32	4.66	4.66	1.49	1.49	5.72	5.72
50 x 50	2 x 2	1.5	0.059	2.24	13.43	0.68	2.85	11.07	11.07	4.43	4.43	1.97	1.97	5.15	5.15
		1.9	0.075	2.80	16.78	0.85	3.56	13.55	13.55	5.42	5.42	1.95	1.95	6.37	6.37
		2.3	0.091	3.34	20.03	1.02	4.25	15.86	15.86	6.34	6.34	1.93	1.93	7.52	7.52
		3.0	0.120	4.25	25.47	1.29	5.41	19.47	19.47	7.79	7.79	1.90	1.90	9.39	9.39
		4.0	0.156	5.45	32.72	1.66	6.95	23.74	23.74	9.49	9.49	1.85	1.85	11.73	11.73
		4.5	0.177	6.02	36.12	1.83	7.67	25.50	25.50	10.20	10.20	1.82	1.82	12.76	12.76
65 x 65	2 1/2 x 2 1/2	1.5	0.059	2.95	17.67	0.90	3.75	25.02	25.02	7.70	7.70	2.58	2.58	8.89	8.89
		1.9	0.075	3.69	22.15	1.13	4.70	30.89	30.89	9.51	9.51	2.56	2.56	11.05	11.05
		2.3	0.091	4.42	26.53	1.35	5.63	36.45	36.45	11.21	11.21	2.54	2.54	13.13	13.13
		3.0	0.120	5.66	33.95	1.72	7.21	45.42	45.42	13.97	13.97	2.51	2.51	16.57	16.57
		4.0	0.156	7.34	44.03	2.24	9.35	56.64	56.64	17.43	17.43	2.46	2.46	21.05	21.05
		4.5	0.177	8.14	48.84	2.48	10.37	61.59	61.59	18.95	18.95	2.44	2.44	23.10	23.10
		5.0	0.197	8.91	53.49	2.72	11.36	66.10	66.10	20.34	20.34	2.41	2.41	25.03	25.03
		6.0	0.236	10.39	62.33	3.17	13.23	73.91	73.91	22.74	22.74	2.36	2.36	28.53	28.53

6c (i) Cold Formed Square Hollow Sections

(American Standard)
(Dimension and Properties in accordance to ASTM A - 500 Grade A and B / Manufacturer's Standard)



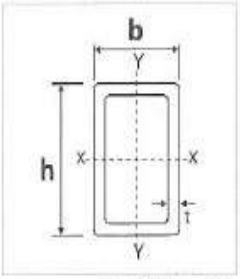
Square Hollow Sections

ASTM A - 500

Nominal Size		Wall Thickness		Calculated Weight			Area	Moment Of Inertia		Section Modulus		Radius Of Gyration		Plastic Modulus	
b x b		t					A	Ix	Iy	zx	zy	ix	iy	Sx	Sy
mm	in	mm	in	Kg/m	Kg/6m	Kg/ft	cm^2	cm^4	cm^4	cm^3	cm^3	cm	cm	cm^3	cm^3
75 x 75	3 x 3	1.5	0.059	3.42	20.50	1.04	4.35	38.92	38.92	10.38	10.38	2.99	2.99	11.94	11.94
		1.9	0.075	4.29	25.73	1.31	5.46	48.23	48.23	12.86	12.86	2.97	2.97	14.89	14.89
		2.3	0.091	5.14	30.86	1.57	6.55	57.10	57.10	15.23	15.23	2.95	2.95	17.74	17.74
		3.0	0.120	6.60	39.60	2.01	8.41	71.62	71.62	19.10	19.10	2.92	2.92	22.49	22.49
		4.0	0.156	8.59	51.56	2.62	10.95	90.19	90.19	24.05	24.05	2.87	2.87	28.76	28.76
		4.5	0.177	9.55	57.31	2.91	12.17	98.55	98.55	26.28	26.28	2.85	2.85	31.68	31.68
		5.0	0.197	10.48	62.91	3.20	13.36	106.33	106.33	28.35	28.35	2.82	2.82	34.46	34.46
		6.0	0.236	12.27	73.63	3.74	15.63	120.16	120.16	32.04	32.04	2.77	2.77	39.58	39.58
90 x 90	3 1/2 x 3 1/2	2.3	0.091	6.23	37.36	1.90	7.93	100.79	100.79	22.40	22.40	3.56	3.56	25.93	25.93
		3.0	0.120	8.01	48.08	2.44	10.21	127.28	127.28	28.29	28.29	3.53	3.53	33.04	33.04
		4.0	0.156	10.48	62.87	3.19	13.35	161.92	161.92	35.98	35.98	3.48	3.48	42.58	42.58
		4.5	0.177	11.67	70.03	3.56	14.87	177.87	177.87	39.53	39.53	3.46	3.46	47.09	47.09
		5.0	0.197	12.84	77.04	3.91	16.36	192.93	192.93	42.87	42.87	3.43	3.43	51.41	51.41
		6.0	0.236	15.10	90.59	4.60	19.23	220.48	220.48	48.99	48.99	3.39	3.39	59.54	59.54
100 x 100	4 x 4	2.3	0.091	6.95	41.69	2.12	8.85	139.73	139.73	27.95	27.95	3.97	3.97	32.26	32.26
		3.0	0.120	8.96	53.73	2.73	11.41	177.05	177.05	35.41	35.41	3.94	3.94	41.21	41.21
		4.0	0.156	11.73	70.40	3.58	14.95	226.35	226.35	45.27	45.27	3.89	3.89	53.30	53.30
		4.5	0.177	13.08	78.51	3.99	16.67	249.29	249.29	49.86	49.86	3.87	3.87	59.04	59.04
		5.0	0.197	14.41	86.46	4.39	18.36	271.10	271.10	54.22	54.22	3.84	3.84	64.59	64.59
		6.0	0.236	16.98	101.89	5.18	21.63	311.47	311.47	62.29	62.29	3.79	3.79	75.10	75.10
125 x 125	5 x 5	3.0	0.120	11.31	67.86	3.45	14.41	354.50	354.50	56.72	56.72	4.96	4.96	65.56	65.56
		4.0	0.156	14.87	89.24	4.53	18.95	457.23	457.23	73.16	73.16	4.91	4.91	85.33	85.33
		4.5	0.177	16.62	99.70	5.06	21.17	505.83	505.83	80.93	80.93	4.89	4.89	94.84	94.84
		5.0	0.197	18.33	110.01	5.59	23.36	552.62	552.62	88.42	88.42	4.86	4.86	104.10	104.10
		6.0	0.236	21.69	130.15	6.61	27.63	640.89	640.89	102.54	102.54	4.82	4.82	121.87	121.87
150 x 150	6 x 6	4.0	0.156	18.01	108.08	5.49	22.95	807.82	807.82	107.71	107.71	5.93	5.93	124.87	124.87
		4.5	0.177	20.15	120.90	6.14	25.67	896.30	896.30	119.51	119.51	5.91	5.91	139.08	139.08
		5.0	0.197	22.26	133.56	6.78	28.36	982.12	982.12	130.95	130.95	5.89	5.89	152.98	152.98
		6.0	0.236	26.40	158.41	8.05	33.63	1145.91	1145.91	152.79	152.79	5.84	5.84	179.88	179.88
		8.0	0.315	33.95	203.67	10.35	43.24	1411.83	1411.83	188.24	188.24	5.71	5.71	225.96	225.96
		9.0	0.354	37.66	225.98	11.48	47.98	1537.39	1537.39	204.99	204.99	5.66	5.66	248.20	248.20
		10.0	0.394	41.26	247.59	12.58	52.57	1652.53	1652.53	220.34	220.34	5.61	5.61	269.17	269.17
		12.0	0.472	47.15	282.88	14.37	60.06	1779.77	1779.77	237.30	237.30	5.44	5.44	297.66	297.66
200 x 200	8 x 8	4.5	0.177	27.21	163.29	8.30	34.67	2191.55	2191.55	219.15	219.15	7.95	7.95	252.86	252.86
		5.0	0.197	30.11	180.66	9.18	38.36	2410.09	2410.09	241.01	241.01	7.93	7.93	278.87	278.87
		6.0	0.236	35.82	214.93	10.92	45.63	2832.75	2832.75	283.27	283.27	7.88	7.88	329.67	329.67
		8.0	0.315	46.51	279.03	14.17	59.24	3566.26	3566.26	356.63	356.63	7.76	7.76	420.86	420.86
		9.0	0.354	51.79	310.76	15.79	65.98	3918.46	3918.46	391.85	391.85	7.71	7.71	465.35	465.35
		10.0	0.394	56.96	341.79	17.36	72.57	4251.06	4251.06	425.11	425.11	7.65	7.65	508.08	508.08
		12.0	0.472	65.99	395.92	20.11	84.06	4730.23	4730.23	473.02	473.02	7.50	7.50	575.61	575.61
250 x 250	10 x 10	5.0	0.197	37.96	227.76	11.57	48.36	4805.01	4805.01	384.40	384.40	9.97	9.97	442.26	442.26
		6.0	0.236	45.24	271.45	13.79	57.63	5672.00	5672.00	453.76	453.76	9.92	9.92	524.45	524.45
		8.0	0.315	59.07	354.39	18.00	75.24	7229.21	7229.21	578.34	578.34	9.80	9.80	675.77	675.77
		9.0	0.354	65.92	395.54	20.09	83.98	7983.76	7983.76	638.70	638.70	9.75	9.75	750.00	750.00
		10.0	0.394	72.66	435.99	22.15	92.57	8706.68	8706.68	696.53	696.53	9.70	9.70	822.00	822.00
		12.0	0.472	84.83	508.96	25.86	108.06	9859.42	9859.42	788.75	788.75	9.55	9.55	943.56	943.56
300 x 300	12 x 12	6.0	0.236	54.66	327.97	16.66	69.63	9963.67	9963.67	664.24	664.24	11.96	11.96	764.23	764.23
		8.0	0.315	71.63	429.75	21.83	91.24	12800.69	12800.69	853.38	853.38	11.84	11.84	990.68	990.68
		9.0	0.354	80.05	480.32	24.40	101.98	14183.29	14183.29	945.55	945.55	11.79	11.79	1102.14	1102.14
		10.0	0.394	88.36	530.19	26.93	112.57	15519.37	15519.37	1034.62	1034.62	11.74	11.74	1210.91	1210.91
		12.0	0.472	103.67	622.00	31.60	132.06	17767.37	17767.37	1184.49	1184.49	11.60	11.60	1401.51	1401.51

6c (ii) Cold Formed Rectangular Hollow Sections

(American Standard)
(Dimension and Properties in accordance to
ASTM A - 500 Grade A and B / Manufacturer's Standard)



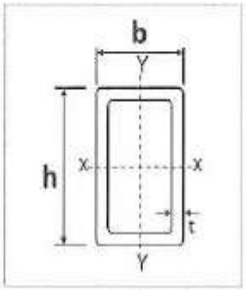
Rectangular Hollow Sections

ASTM A - 500

Nominal Size		Wall Thickness		Calculated Weight			Area	Moment Of Inertia		Section Modulus		Radius Of Gyration		Plastic Modulus	
h x b		t					A	Ix	Iy	zx	zy	Ix	Iy	Sx	Sy
mm	in	mm	in	Kg/m	Kg/6m	Kg/ft	cm²	cm⁴	cm⁴	cm³	cm³	cm	cm	cm³	cm³
38 x 19	1 1/2 x 1 3/4	1.5	0.059	1.23	7.36	0.37	1.56	2.77	0.93	1.46	0.98	1.33	0.77	1.85	1.14
50 x 25	2 x 1	1.5	0.059	1.65	9.90	0.50	2.10	6.65	2.25	2.66	1.80	1.78	1.04	3.33	2.05
		1.9	0.075	2.05	12.31	0.63	2.61	8.06	2.71	3.22	2.16	1.76	1.02	4.08	2.51
		2.3	0.091	2.44	14.61	0.74	3.10	9.31	3.10	3.72	2.48	1.73	1.00	4.78	2.92
		3.0	0.120	3.07	18.41	0.94	3.91	11.17	3.67	4.47	2.93	1.69	0.97	5.86	3.56
65 x 35		1.5	0.059	2.24	13.43	0.68	2.85	15.94	6.12	4.91	3.50	2.36	1.47	6.03	3.93
		1.9	0.075	2.80	16.78	0.85	3.56	19.54	7.46	6.01	4.26	2.34	1.45	7.46	4.85
		2.3	0.091	3.34	20.03	1.02	4.25	22.88	8.69	7.04	4.96	2.32	1.43	8.81	5.72
		3.0	0.120	4.25	25.47	1.29	5.41	28.10	10.57	8.65	6.04	2.28	1.40	10.99	7.11
		4.0	0.156	5.45	32.72	1.66	6.95	34.28	12.72	10.55	7.27	2.22	1.35	13.73	8.83
65 x 38	2 1/2 x 1 1/2	1.5	0.059	2.31	13.86	0.70	2.94	16.85	7.37	5.18	3.88	2.39	1.58	6.32	4.37
		1.9	0.075	2.89	17.32	0.88	3.68	20.68	9.00	6.36	4.74	2.37	1.56	7.82	5.40
		2.3	0.091	3.45	20.68	1.05	4.39	24.23	10.50	7.46	5.53	2.35	1.55	9.24	6.37
		3.0	0.120	4.39	26.32	1.34	5.59	29.83	12.83	9.18	6.75	2.31	1.51	11.55	7.93
		4.0	0.156	5.64	33.86	1.72	7.19	36.52	15.53	11.24	8.17	2.25	1.47	14.46	9.89
75 x 25	3 x 1	1.5	0.059	2.24	13.43	0.68	2.85	18.66	3.29	4.98	2.63	2.56	1.07	6.43	2.94
		1.9	0.075	2.80	16.78	0.85	3.56	22.84	3.98	6.09	3.18	2.53	1.06	7.94	3.61
		2.3	0.091	3.34	20.03	1.02	4.25	26.70	4.59	7.12	3.67	2.51	1.04	9.37	4.23
		3.0	0.120	4.25	25.47	1.29	5.41	32.72	5.49	8.72	4.39	2.46	1.01	11.69	5.21
75 x 38	3 x 1 1/2	1.5	0.059	2.55	15.27	0.78	3.24	23.93	8.37	6.38	4.40	2.72	1.61	7.86	4.92
		1.9	0.075	3.18	19.11	0.97	4.06	29.44	10.24	7.85	5.39	2.69	1.59	9.75	6.08
		2.3	0.091	3.81	22.84	1.16	4.85	34.61	11.97	9.23	6.30	2.67	1.57	11.55	7.19
		3.0	0.120	4.86	29.15	1.48	6.19	42.83	14.67	11.42	7.72	2.63	1.54	14.49	8.98
		4.0	0.156	6.27	37.62	1.91	7.99	52.84	17.85	14.09	9.40	2.57	1.50	18.26	11.25
		4.5	0.177	6.94	41.63	2.11	8.84	57.12	19.17	15.23	10.09	2.54	1.47	19.95	12.25
75 x 50	3 x 2	1.5	0.059	2.83	16.97	0.86	3.60	28.79	15.48	7.68	6.19	2.83	2.07	9.18	6.97
		1.9	0.075	3.54	21.25	1.08	4.51	35.54	19.05	9.48	7.62	2.81	2.05	11.42	8.65
		2.3	0.091	4.24	25.44	1.29	5.40	41.90	22.41	11.17	8.96	2.79	2.04	13.56	10.26
		3.0	0.120	5.42	32.54	1.65	6.91	52.17	27.76	13.91	11.10	2.75	2.00	17.09	12.91
		4.0	0.156	7.02	42.14	2.14	8.95	64.96	34.34	17.32	13.74	2.69	1.96	21.66	16.33
		4.5	0.177	7.79	46.72	2.37	9.92	70.56	37.19	18.82	14.87	2.67	1.94	23.75	17.88
		5.0	0.197	8.52	51.13	2.60	10.86	75.65	39.75	20.17	15.90	2.64	1.91	25.71	19.33
100 x 50	4 x 2	1.5	0.059	3.42	20.50	1.04	4.35	57.77	19.89	11.55	7.96	3.64	2.14	14.16	8.79
		1.9	0.075	4.29	25.73	1.31	5.46	71.62	24.55	14.32	9.82	3.62	2.12	17.65	10.94
		2.3	0.091	5.14	30.86	1.57	6.55	84.83	28.95	16.97	11.58	3.60	2.10	21.03	13.01
		3.0	0.120	6.60	39.60	2.01	8.41	106.46	36.06	21.29	14.42	3.56	2.07	26.66	16.44
		4.0	0.156	8.59	51.56	2.62	10.95	134.14	44.95	26.83	17.98	3.50	2.03	34.10	20.93
		4.5	0.177	9.55	57.31	2.91	12.17	146.61	48.87	29.32	19.55	3.47	2.00	37.56	23.00
		5.0	0.197	10.48	62.91	3.20	13.36	158.19	52.45	31.64	20.98	3.44	1.98	40.84	24.95
		6.0	0.236	12.27	73.63	3.74	15.63	178.75	58.67	35.75	23.47	3.38	1.94	46.90	28.52
125 x 50	5 x 2	2.3	0.091	6.05	36.28	1.84	7.70	148.24	35.50	23.72	14.20	4.39	2.15	29.94	15.75
		3.0	0.120	7.78	46.67	2.37	9.91	187.02	44.35	29.92	17.74	4.34	2.12	38.11	19.96
		4.0	0.156	10.16	60.98	3.10	12.95	237.53	55.56	38.01	22.22	4.28	2.07	49.03	25.53
		4.5	0.177	11.32	67.91	3.45	14.42	260.68	60.55	41.71	24.22	4.25	2.05	54.17	28.12
		5.0	0.197	12.45	74.68	3.79	15.86	282.46	65.16	45.19	26.07	4.22	2.03	59.10	30.58
		6.0	0.236	14.63	87.76	4.46	18.63	322.00	73.28	51.52	29.31	4.16	1.98	68.32	35.12
150 x 50	6 x 2	2.3	0.091	6.95	41.69	2.12	8.85	235.71	42.04	31.43	16.82	5.16	2.18	40.28	18.49
		3.0	0.120	8.96	53.73	2.73	11.41	298.55	52.65	39.81	21.06	5.12	2.15	51.43	23.49
		4.0	0.156	11.73	70.40	3.58	14.95	381.39	66.16	50.85	26.47	5.05	2.10	66.47	30.13
		4.5	0.177	13.08	78.51	3.99	16.67	419.82	72.24	55.98	28.89	5.02	2.08	73.60	33.24
		5.0	0.197	14.41	86.46	4.39	18.36	456.29	77.87	60.84	31.15	4.99	2.06	80.48	36.20
		6.0	0.236	16.98	101.89	5.18	21.63	523.47	87.89	69.80	35.16	4.92	2.02	93.48	41.72

6 c (ii) Cold Formed Rectangular Hollow Sections

(American Standard)
(Dimension and Properties in accordance to
ASTM A - 500 Grade A and B / Manufacturer's Standard)



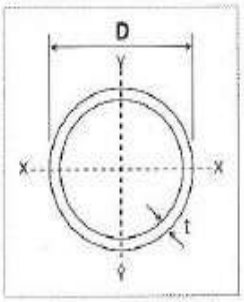
Rectangular Hollow Sections

ASTM A - 500

Nominal Size		Wall Thickness		Calculated Weight			Area	Moment Of Inertia		Section Modulus		Radius Of Gyration		Plastic Modulus	
h x b		t					A	Ix	Iy	zx	zy	ix	iy	Sx	Sy
mm	in	mm	in	Kg/m	Kg/6m	Kg/ft	cm^2	cm^4	cm^4	cm^3	cm^3	cm	cm	cm^3	cm^3
100 x 75	4 x 3	2.3	0.091	6.05	36.28	1.84	7.70	112.28	72.30	22.46	19.28	3.82	3.06	26.64	21.92
		3.0	0.120	7.78	46.67	2.37	9.91	141.75	91.07	28.35	24.29	3.78	3.03	33.93	27.89
		4.0	0.156	10.16	60.98	3.10	12.95	180.25	115.42	36.05	30.78	3.73	2.99	43.70	35.86
		4.5	0.177	11.32	67.91	3.45	14.42	197.95	126.55	39.59	33.75	3.71	2.96	48.30	39.61
		5.0	0.197	12.45	74.68	3.79	15.86	214.64	137.00	42.93	36.53	3.68	2.94	52.72	43.21
		6.0	0.236	14.63	87.76	4.46	18.63	245.11	155.96	49.02	41.59	3.63	2.89	61.00	49.93
125 x 75	5 x 3	2.3	0.091	6.95	41.69	2.12	8.85	191.52	87.51	30.64	23.33	4.65	3.14	36.99	26.10
		3.0	0.120	8.96	53.73	2.73	11.41	242.85	110.52	38.86	29.47	4.61	3.11	47.26	33.29
		4.0	0.156	11.73	70.40	3.58	14.95	310.76	140.65	49.72	37.51	4.56	3.07	61.13	42.96
		4.5	0.177	13.08	78.51	3.99	16.67	342.40	154.54	54.78	41.21	4.53	3.04	67.73	47.55
		5.0	0.197	14.41	86.46	4.39	18.36	372.51	167.68	59.60	44.71	4.50	3.02	74.10	51.96
		6.0	0.236	16.98	101.89	5.18	21.63	428.29	191.76	68.53	51.14	4.45	2.98	86.17	60.28
150 x 75	6 x 3	2.3	0.091	7.85	47.11	2.39	10.00	298.43	102.71	39.79	27.39	5.46	3.20	48.77	30.28
		3.0	0.120	10.13	60.80	3.09	12.91	379.59	129.97	50.61	34.66	5.42	3.17	62.45	38.69
		4.0	0.156	13.30	79.82	4.06	16.95	488.00	165.88	65.07	44.24	5.37	3.13	81.07	50.06
		4.5	0.177	14.85	89.11	4.53	18.92	538.94	182.54	71.86	48.68	5.34	3.11	89.97	55.48
		5.0	0.197	16.37	98.23	4.99	20.86	587.74	198.36	78.37	52.90	5.31	3.08	98.61	60.71
		6.0	0.236	19.34	116.02	5.89	24.63	679.08	227.56	90.54	60.68	5.25	3.04	115.08	70.63
150 x 100	6 x 4	3.0	0.120	11.31	67.86	3.45	14.41	460.64	247.64	61.42	49.53	5.65	4.15	73.48	55.76
		4.0	0.156	14.87	89.24	4.53	18.95	594.60	318.57	79.28	63.71	5.60	4.10	95.67	72.50
		4.5	0.177	16.62	99.70	5.06	21.17	658.06	351.96	87.74	70.39	5.58	4.08	106.34	80.53
		5.0	0.197	18.33	110.01	5.59	23.36	719.20	384.02	95.89	76.80	5.55	4.05	116.73	88.34
		6.0	0.236	21.69	130.15	6.61	27.63	834.69	444.19	111.29	88.84	5.50	4.01	136.68	103.30
200 x 100	8 x 4	4.0	0.156	18.01	108.08	5.49	22.95	1199.71	410.78	119.97	82.16	7.23	4.23	148.04	91.70
		4.5	0.177	20.15	120.90	6.14	25.67	1331.44	454.64	133.14	90.93	7.20	4.21	164.89	102.02
		5.0	0.197	22.26	133.56	6.78	28.36	1459.26	496.94	145.93	99.39	7.17	4.19	181.37	112.09
		6.0	0.236	26.40	158.41	8.05	33.63	1703.31	576.91	170.33	115.38	7.12	4.14	213.27	131.50
		8.0	0.315	33.95	203.67	10.35	43.24	2090.84	705.36	209.08	141.07	6.95	4.04	267.26	164.65
		9.0	0.354	37.66	225.98	11.48	47.98	2275.60	764.43	227.56	152.89	6.89	3.99	293.45	180.46
		10.0	0.394	41.26	247.59	12.58	52.57	2444.40	817.74	244.44	163.55	6.82	3.94	318.08	195.25
		12.0	0.472	47.15	282.88	14.37	60.06	2606.71	875.58	260.67	175.12	6.59	3.82	350.01	215.32
200 x 150	8 x 6	4.0	0.156	21.15	126.92	6.45	26.95	1583.92	1021.03	158.39	136.14	7.67	6.16	187.24	154.07
		4.5	0.177	23.68	142.09	7.22	30.17	1761.49	1134.54	176.15	151.27	7.64	6.13	208.87	171.82
		5.0	0.197	26.18	157.11	7.98	33.36	1934.67	1245.04	193.47	166.00	7.62	6.11	230.12	189.23
		6.0	0.236	31.11	186.67	9.48	39.63	2268.03	1457.13	226.80	194.28	7.56	6.06	271.47	223.08
		8.0	0.315	40.23	241.35	12.26	51.24	2828.55	1815.54	282.85	242.07	7.43	5.95	344.06	282.76
		9.0	0.354	44.73	268.37	13.63	56.98	3097.03	1985.32	309.70	264.71	7.37	5.90	379.40	311.65
		10.0	0.394	49.11	294.69	14.97	62.57	3347.73	2143.36	334.77	285.78	7.31	5.85	413.08	339.17
		12.0	0.472	56.57	339.40	17.24	72.06	3668.47	2352.53	366.85	313.67	7.14	5.71	462.81	380.46
250 x 150	10 x 6	4.5	0.177	27.21	163.29	8.30	34.67	3003.79	1372.78	240.30	183.04	9.31	6.29	289.92	204.55
		5.0	0.197	30.11	180.66	9.18	38.36	3304.18	1507.95	264.33	201.06	9.28	6.27	319.76	225.48
		6.0	0.236	35.82	214.93	10.92	45.63	3885.56	1768.35	310.85	235.78	9.23	6.23	378.05	266.28
		8.0	0.315	46.51	279.03	14.17	59.24	4885.79	2219.25	390.86	295.90	9.08	6.12	482.17	339.56
		9.0	0.354	51.79	310.76	15.79	65.98	5368.90	2433.25	429.51	324.43	9.02	6.07	533.10	375.10
		10.0	0.394	56.96	341.79	17.36	72.57	5825.01	2634.20	466.00	351.23	8.96	6.02	582.00	409.17
		12.0	0.472	65.99	395.92	20.11	84.06	6457.90	2925.29	516.63	390.04	8.77	5.90	657.96	463.26
300 x 200	12 x 8	5.0	0.197	37.96	227.76	11.57	48.36	6241.05	3360.92	416.07	336.09	11.36	8.34	495.65	376.37
		6.0	0.236	45.24	271.45	13.79	57.63	7370.23	3962.19	491.35	396.22	11.31	8.29	587.83	446.07
		8.0	0.315	59.07	354.39	18.00	75.24	9389.28	5041.67	625.95	504.17	11.17	8.19	757.08	574.46
		9.0	0.354	65.92	395.54	20.09	83.98	10371.43	5561.32	691.43	556.13	11.11	8.14	840.24	637.25
		10.0	0.394	72.66	435.99	22.15	92.57	11312.71	6057.73	754.18	605.77	11.05	8.09	920.91	698.08
		12.0	0.472	84.83	508.96	25.86	108.06	12787.85	6853.75	852.52	685.37	10.88	7.96	1055.91	801.21

6c (i) Cold Formed Circular Hollow Sections

(Australian / New Zealand Standard)
(Dimensions and Properties in accordance to AS / NZS 1163 : 2009)



Circular Hollow Sections

AS/NZS 1163 : 2009

Designation			Unit Weight	External Surface Area		Ratio	Gross Area Of Cross-section	About Any Axis				Torsion Constant	Torsion Modulus
Outside Diameter	x	Thickness		Per Unit Length	Per Unit Mass			Second Moment Of Area	Elastic Section Modulus	Plastic Section Modulus	Radius Of Gyration		
D	x	t	m			D/t	A_g	I	Z	S	r	J	C
mm	x	mm	kg/m	m^2/m	m^2/t		mm^2	$10^6\ mm^4$	$10^3\ mm^3$	$10^3\ mm^3$	mm	$10^3\ mm^4$	$10^3\ mm^3$
114.3	x	3.2	8.77	0.359	41.0	35.7	1120	1.72	30.2	39.5	39.3	3.45	60.4
114.3	x	3.6	9.83	0.359	36.5	31.8	1250	1.92	33.6	44.1	39.2	3.84	67.2
114.3	x	4.5	12.2	0.359	29.5	25.4	1550	2.34	41.0	54.3	38.9	4.69	82.0
114.3	x	4.8	13.0	0.359	27.7	23.8	1650	2.48	43.4	57.6	38.8	4.96	86.6
114.3	x	5.4	14.5	0.359	24.8	21.2	1850	2.75	48.0	64.1	38.5	5.49	96.1
114.3	x	6.0	16.0	0.359	22.4	19.1	2040	3.00	52.5	70.4	38.3	6.00	105
139.7	x	3.0	10.1	0.439	43.4	46.6	1290	3.01	43.1	56.1	48.3	6.02	86.2
139.7	x	3.5	11.8	0.439	37.3	39.9	1500	3.47	49.7	64.9	48.2	6.95	99.5
139.7	x	5.0	16.6	0.439	26.4	27.9	2120	4.81	68.8	90.8	47.7	9.61	138
139.7	x	5.4	17.9	0.439	24.5	25.9	2280	5.14	73.7	97.4	47.5	10.3	147
165.1	x	3.0	12.0	0.519	43.2	55.0	1530	5.02	60.8	78.8	57.3	10.0	122
165.1	x	3.5	13.9	0.519	37.2	47.2	1780	5.80	70.3	91.4	57.1	11.6	141
165.1	x	5.0	19.7	0.519	26.3	33.0	2510	8.07	97.7	128	56.6	16.1	195
165.1	x	5.4	21.3	0.519	24.4	30.6	2710	8.65	105	138	56.5	17.3	209
168.3	x	4.8	19.4	0.529	27.3	35.1	2470	8.25	98.0	128	57.8	16.5	196
168.3	x	6.4	25.6	0.529	20.7	26.3	3260	10.7	127	168	57.3	21.4	254
168.3	x	7.1	28.2	0.529	18.7	23.7	3600	11.7	139	185	57.0	23.4	278
219.1	x	4.8	25.4	0.688	27.1	45.6	3230	18.6	169	220	75.8	37.1	339
219.1	x	6.4	33.6	0.688	20.5	34.2	4280	24.2	221	290	75.2	48.4	442
219.1	x	8.2	42.6	0.688	16.1	26.7	5430	30.3	276	365	74.6	60.5	552
273.1	x	4.8	31.8	0.858	27.0	56.9	4050	36.4	267	346	94.9	72.8	533
273.1	x	6.4	42.1	0.858	20.4	42.7	5360	47.7	349	455	94.3	95.4	699
273.1	x	9.3	60.5	0.858	14.2	29.4	7710	67.1	492	647	93.3	134	983
323.9	x	2.7	97.5	1.02	10.4	25.5	12400	151	930	1230	110	301	1860
323.9	x	6.4	50.1	1.02	20.3	50.6	6380	80.5	497	645	112	161	994
323.9	x	9.5	73.7	1.02	13.8	34.1	9380	116	717	939	111	232	1430
355.6	x	6.4	55.1	1.12	20.3	55.6	7020	107	602	781	123	214	1200
355.6	x	9.5	81.1	1.12	13.8	37.4	10300	155	871	1140	122	310	1740
355.6	x	12.7	107	1.12	10.4	28.0	13700	201	1130	1490	121	403	2260
406.4	x	6.4	63.1	1.28	20.2	63.5	8040	161	792	1020	141	322	1580
406.4	x	9.5	93.0	1.28	13.7	42.8	11800	233	1150	1500	140	467	2300
406.4	x	12.7	123	1.28	10.4	32.0	15700	305	1500	1970	139	609	3000
457.0	x	6.4	71.1	1.44	20.2	71.4	9060	230	1010	1300	159	460	2010
457.0	x	9.5	105	1.44	13.7	48.1	13400	334	1460	1900	158	669	2930
457.0	x	12.7	139	1.44	10.3	36.0	17700	438	1920	2510	157	876	3830
508.0	x	6.4	79.2	1.60	20.2	79.4	10100	317	1250	1610	177	634	2500
508.0	x	9.5	117	1.60	13.7	53.5	14900	462	1820	2360	176	925	3640
508.0	x	12.7	155	1.60	10.3	40.0	19800	606	2390	3120	175	1210	4770
610.0	x	6.4	95.3	1.92	20.1	95.3	12100	553	1810	2330	213	1110	3620
610.0	x	9.5	141	1.92	13.6	64.2	17900	808	2850	3430	212	1620	5300
610.0	x	12.7	187	1.92	10.2	48.0	23800	1060	3490	4530	211	2130	6970

6c) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel

Structural Tubing in Round and Shapes – America Standard (Extracts from ASTM A-500 : 2007)

General Information

ASTM A-500 covers cold-formed welded carbon steel round, square, rectangular, or special shape structural tubing for welded, riveted, or bolted construction of bridges and buildings, and for general structural purposes.

Chemical Composition

The Chemical analysis shall conform to the limits given in Table 1 for the appropriate grade.

Table 1. Chemical Requirements

Grades		Chemical Composition %				
		C max.	Mn max.	P max.	S max.	Cu min.
Grade A, B and D	Heat Analysis	0.26	1.35	0.035	0.035	0.20
	Product Analysis	0.30	1.40	0.045	0.045	0.18
Grade C	Heat Analysis	0.23	1.35	0.035	0.035	0.20
	Product Analysis	0.27	1.40	0.045	0.045	0.18

Mechanical Strength (Tensile Test)

For Tensile test. The yield strength, tensile strength and elongation of the test piece shall conform to the limits given in Table 2 for the appropriate grade.

Table 2. Mechanical Properties

Shaped Structural Tubing			
Grade	Tensile Strength, min, psi (MPa)	Yield Strength, min, psi (MPa)	Elongation in 2 in. (50 mm) min %
Grade A	45000 (310)	39000 (270)	25
Grade B	58000 (400)	46000 (315)	23
Grade C	62000 (425)	50000 (345)	21
Grade D	58000 (400)	36000 (250)	23

Round Structural Tubing			
Grade	Tensile Strength, min, psi (MPa)	Yield Strength, min, psi (MPa)	Elongation in 2 in. (50 mm) min %
Grade A	45000 (310)	33000 (230)	25
Grade B	58000 (400)	42000 (290)	23
Grade C	62000 (425)	46000 (315)	21
Grade D	58000 (400)	36000 (250)	23

Cold Flattening Test

The flattening test shall be made on round structural tubing. A flattening test is not required for shaped structural tubing. For welded round structural tubing, a test specimen at least 4 in. [100mm] in length shall be flattened cold between parallel plates in three steps with the weld located either 0° or 90° from the line of direction of the force.

1) During the first step, which is a test for ductility of the weld, no cracks or breaks on the inside or outside surfaces of the test specimen shall be present until the distance between the plates is less than two-thirds of the specified outside diameter of the tubing.

2) For the second step, no cracks or breaks on the inside or outside parent metal surfaces of the test specimen, shall be present until the distance between the plates is less than one-half of the specified outside diameter of the tubing.

3) During the third step, which is a test for soundness, the flattening shall be continued until the test specimen breaks or the opposite walls of the test specimen meet. Evidence of laminated or unsound material or of incomplete weld that is revealed by the flattening test shall be cause for rejection.

Tolerances On Dimensions and Mass

The tolerances on dimensions shall respectively conform to Table 3.

Table 3. Tolerance on dimensions

Characteristic	Circular Hollow Sections		Square And Rectangular Hollow Sections	
Outside Dimension (*OD*)	Specified OD (mm)	Tolerance (mm)	Specified Outside Large Flat Dimension in. (mm)	Tolerances, plus & minus in. (mm)
	OD ≤ 48	± 0.50 %	2 1/2 (65) or under	0.020 (0.50)
			Over 2 1/2 to 3 1/2 (65 to 90), incl	0.025 (0.60)
	OD ≥ 50	± 0.75 %	Over 3 1/2 to 5 1/2 (90 to 140), incl	0.030 (0.80)
			Over 5 1/2 (140)	0.01 times large flat dimension
Thickness (*t*)	± 10 % of nominal thickness, exclusive of weld area			
Length	Length tolerance for specific lengths, in. (mm)	22 ft (6.5 m) And Under		Over 22 ft (6.5 m)
		Over	Under	Over	Under
		1/2 (13)	1/4 (6)	3/4 (19)	1/4 (6)
Straightness	$1/8 \text{ in} \times \frac{\text{Total Length in ft}}{5}$			
Out-of-roundness	-			
Concavity / convexity	Shall be not exceed the outside dimension tolerance			
Radius of Corners	-	Shall not exceed three times the specified wall thickness		
Squareness of Sides	-	adjacent sides shall be square (90°), with permissible variation of ± 2° max.		
Twist	-	Specified Dia. of longest side, in (mm)	Maximum Permissible Variations in Twist per 3ft of length (Twist per meter of length)	
			in.	mm
		1 1/2 (40) and under	0.050	1.3
		Over 1 1/2 to 2 1/2 (40 to 65), incl	0.062	1.6
		Over 21 1/2 to 4 (65 to 100), incl	0.075	1.9
		Over 4 to 6 (100 to 150). Incl	0.087	2.2
		Over 6 to 8 (150 to 200), incl	0.100	2.5
		Over 8 (200)	0.112	2.8
Inner Flash	-			
End tolerance on diameter	-			
End Facing	-			
Mass (*m*) per unit length	-			